

MAIN AND PERISTALTIC DATSHEETS

NKP-DC-S10B, Peristaltic

<https://www.kamoer.cn/us/product/params.html?id=4004> Datasheet

- 800 Hour lifetime
- ~\$16 x 3/2 = \$24
- <12 watts, 4-14 mL/min (enough for nutrients)
- [purchase link amazon](#)

NOTE: BE SURE TO TEST SPECS OF CURRENT WATER PUMP

DEFENSIVE ELECTRICAL UPDATES

The prior system had no built-in defensive mechanisms and fault checks. Update will include

UPDATES

- 1) Shunt sensing: Built into main pump lines, program arduino to measure voltage/shut down FET if detecting anomaly
- 2) Red LEDs: When voltage is not at correct level, Arduino will output red warning light
- 3) Flyback diodes: Suppress voltage transients from pump inductance

PUMP SENSOR COUNT REDUCTION

Water level sensors should be simplified: Unnecessary to have multiple water level sensors if it'll all be flowing anyways.

NOTE: MAKE SURE TO CONSIDER BLOCKAGES FOR WATER LEVEL SENSING

Only 1 needed at the top level, should be enough as long as structures account for drainage

UPDATES

- 1) Decreased water level sensing from 4 -> 1

PDN UPDATES

NOTE: Arduino nano takes in 7-12V, NOT 5V at Vin, allows for approx 0.5V dV drop, make sure Ztarget of PDN < 0.5V/transient

UPDATES

- 1) Decoupling network capacitances update
- 2) 5V buck regulator -> 7V buck regulator Vin
- 3) MBR745G -> MBR1060G diode for both flyback + buck regulator

<https://docs.arduino.cc/hardware/nano/#tech-specs> Nano datasheet

ELECTRICAL TO DO

- 1) Test maximum transients (with flyback included) for pumps, check if within FET tolerance
- 2) Structures team, update drainage system (due to water level sensor removal)
- 3) Measure impedance on all power traces, ensure power drop is within tolerance
- 4) Validate whether thermals will be an issue for PCB (specifically FET/diode/traces)

FIRMWARE TO DO

- 1) Organize files, basic EC + pH + water -> pumps control
- 2) Arduino program that detects voltage/current across shunts -> red LED
- 3) Overflow loop counter to defend against pump constantly pumping
- 4) Median value of ~5-10s of inputs in firmware to filter out analog signal noise

FUTURE UPDATES

- 1) Decentralize defense away from arduino (ex fuses, breakers)
- 2) Update if tolerances specifications are not met (V, Z, I, etc)